

The source sign of the edge Schur scalar

Woodbury and the Sylvester chart are exact; $\tau \rightarrow 0$ is a census; the rate is open

Stefan Hamann

August 29, 2026

Abstract

After rounds 397–416 the bulk half of $P1$ rests (window-emergent defect; no pre-declared exceptional vector). The remaining internal lever is the edge sign $\text{sch} < 0$: an $O(1)$ object in the r401 3×3 Sylvester chart, never previously attacked from the source. Two closed formulae for sch are finite SATZ: the Woodbury identity $\text{sch} = \text{den} - 2 + s^T(A_0 + U_{\text{CD}}U_{\text{CD}}^T)^{-1}s$ and the unnormalized chart $\text{sch} = \varphi_{bb} + \tilde{a}^2 - \tilde{b}^2$ (P1) / $\text{sch} = \varphi_{bb} - (\tilde{a}^2 + \tilde{b}^2)$ (vacuous). Normalized $\tau = (a, b)$ cannot see the sign (d_3 quotients it out). On the named census $\tau \rightarrow 0$ with $|\text{sch}|$ flat of order 0.04, dead χ sit on/across the $\text{sch} = 0$ curve, and the observed rate $|\tau| \sim N^{-0.47}$ is not derived from Uvarov quotients or fold telescoping. No cofinal k_0 . No RH claim.

Status. Lemmas 1–2 are proved (finite identities over $\mathbb{Q}$). Propositions 1–3 are finite machine checks. Ausgang CHART_SCH_EXACT / LIMIT_CENSUS / RATE_OPEN. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The scalar

Write $A_0 = R_{N-3} - \frac{1}{2}I$ on the hole nodes Y , U_{CD} the last two dual Christoffel–Darboux columns, and s the Sherman–Morrison border vector of r362/r401, with border scale $\text{den} > 0$. The mixed 3×3 is

$$\Phi = \begin{pmatrix} K_2 & c_0 \\ c_0^T & \varphi_{bb} \end{pmatrix}, \quad K_2 = I + U_{\text{CD}}^T A_0^{-1} U_{\text{CD}},$$

$c_0 = U_{\text{CD}}^T A_0^{-1} s$ and $\varphi_{bb} = \text{den} - 2 + s^T A_0^{-1} s$. The Schur complement of K_2 in Φ is the scalar $\text{sch} = \varphi_{bb} - c_0^T K_2^{-1} c_0$. Haynsworth (r401) converts the border shift $\text{ind}_- \Phi = 1 + \text{ind}_- A_0$ into $\text{sch} < 0$.

After the 2×2 Sylvester map D_2 of K_2 , $(\tilde{a}, \tilde{b})^T = D_2^T c_0$ are the *unnormalized* couplings; the r401 normalized $\tau = (a, b)$ is $(\tilde{a}, \tilde{b})/\sqrt{|\text{sch}|}$. The normalized family Φ_{edge} has Schur complement $\text{sign}(\text{sch})$ identically, so the sign is a chart choice, not a region of the (a, b) disk.

2 Finite identities

Lemma 1 (Woodbury).

$$\text{sch} = \text{den} - 2 + s^T(A_0 + U_{\text{CD}}U_{\text{CD}}^T)^{-1}s.$$

On the rational $P1$ toy $A_0 = \text{diag}(-\frac{1}{2}, 2, 3)$, $s = e_1$, $\text{den} = 1$ one has $\text{sch} = -\frac{2}{3}$ exactly. On the vacuous toy $A_0 = \text{diag}(2, 3, -1, 4)$, $s = e_0$, $\text{den} = \frac{1}{2}$ one has $\text{sch} = -\frac{7}{6}$.

Lemma 2 (Sylvester chart). *Let $J_2 = \text{diag}(\text{sign } \mu(K_2))$. On the P1 chart $J_2 = \text{diag}(-1, 1)$,*

$$\text{sch} = \varphi_{bb} + \tilde{a}^2 - \tilde{b}^2. \quad (1)$$

On the vacuous chart $J_2 = I_2$,

$$\text{sch} = \varphi_{bb} - (\tilde{a}^2 + \tilde{b}^2). \quad (2)$$

Both identities agree with Lemma 1 over $\mathbb{Q}$ (P1: $\varphi_{bb} = -\frac{1}{2}$, $\tilde{b}^2 = \frac{1}{6}$; vacuous: $\varphi_{bb} = -1$).

Remark 1 (who carries the sign). Write $Q = s^T(A_0 + U_{\text{CD}}U_{\text{CD}}^T)^{-1}s$. Then $\text{sch} = Q - (2 - \text{den})$. On the named census $Q \in [0.283, 0.502]$ and $\text{den} \in [1.460, 1.652]$ are both $O(1)$; sch is their difference, of order 0.04, not the $O(1)$ K_2 -excess of r375. On the vacuous chart with $\varphi_{bb} < 0$, (2) forces $\text{sch} < 0$ for every coupling. On the P1 chart the danger axis is large $|\tilde{a}|$ (the negative K_2 direction adds $+\tilde{a}^2$ to sch).

3 Machine checks

Proposition 1 (sign map). *On the unnormalized disk of radius 0.4 at $\varphi_{bb} = -0.05$, the vacuous chart has $\text{sch} < 0$ at 1257/1257 grid points; the P1 chart only at 989/1257 (hyperbola $\tilde{a}^2 - \tilde{b}^2 = -\varphi_{bb}$). The w9-un disk of radius 0.14 about the origin lies entirely in $\text{sch} < 0$ (minimum $|\text{sch}| = 0.066$). Dropping the border column kills the shift ($\text{ind}_- K_2 = \text{ind}_- A_0$).*

Proposition 2 (flagship and false chart). *At $k_z = 9$, $\text{sch} = -0.06696$, $(\tilde{a}, \tilde{b}) = (-0.13779, -0.01786)$, $\varphi_{bb} = -0.08562$, $Q = 0.33193$, Euclidean distance to the $\text{sch} = 0$ curve 0.241. Woodbury and chart residuals are $2.8 \cdot 10^{-17}$. The false (vacuous) formula on the P1 chart misses by $2\tilde{a}^2 = 0.03797$. The r377 product $h_N h_{N+1}$ and $\det K_2$ are both negative — on the living $\text{nneg} = 1$ branch both coordinates fire; they are not globally equivalent.*

Proposition 3 (census, trend, kills). *MAIN 42/42 have $\text{sch} < 0$ (P1 28, vacuous 14), chart residual $\leq 10^{-17}$. Vacuous windows have $\det K_2 > 0$: $\text{sch} < 0$ is not the r377 alternation. $\log |\tau| \sim \log N$ slope -0.41 (falling); $\log |\text{sch}| \sim \log N$ slope $+0.09$ (flat). EXT six rows, including $k_z = 119$ at $N = 1119$ with $|\tau| = 0.075$, all $\text{sch} < 0$ and $\varphi_{bb} < 0$. Selected anchors $a_k = 2^k$: $k = 5$ ($k_z = 17$) $\text{sch} = -0.109$; $k = 9$ ($k_z = 116$, $N = 1433$) $|\tau| = 0.125$, $\text{sch} = -0.038$. Living χ 37 + 41 have $\text{sch} < 0$; dead χ 5 + 1 have $\text{sch} > 0$, median distance 0.011 against live 0.180, and 3/5 of dead χ_3 escape the living disk $R = 3.2$, while dead P1 keeps $\det K_2 < 0$. Scramble $\text{nneg}(A_0) = 21$; two-period $S = 21$ has $\text{nneg} = 4$.*

4 What is a theorem, what is a census

Lemma 1 and Lemma 2 are theorems. The reduction is: on the vacuous chart, $\varphi_{bb} < 0$ implies $\text{sch} < 0$ outright; on the P1 chart, $\tau_{\text{un}} \rightarrow 0$ plus a uniform $\varphi_{bb} \leq -\delta < 0$ would give $\text{sch} < 0$ cofinally. The census supports $\tau \rightarrow 0$, $|\text{sch}|$ flat of order 0.04, and $\varphi_{bb} < 0$ at large N , two-sided against dead χ . The observed rate $|\tau| \sim N^{-0.47}$ is *not* a corollary of the r392 Uvarov kernel update nor of fold telescoping: the identities exist as exact finite-rank updates of the CD kernel, but they do not force the slope (gate RATE_OPEN; PNT-free in form — the sums are window-finite — and not RH-near). No legitimate k_0 , so $\text{sch}_k < 0$ cofinally on the selected sequence is not proved. Normalized $\tau_\infty = 0$ does not evaluate $\text{sch}(\tau_\infty)$: the scale was quotiented. The $O(1)$ K_2 -excess of r375 is a different coordinate. No RH claim.